

Anonymous Linear Bandits for Multi-User Systems

Mohammad Reza Badri
mohammadreza.badri@epfl.ch
EPFL
Lausanne, Switzerland

Samira Hossein Ghorban
s.hosseinghorban@ipm.ir
Institute for Research in Fundamental Sciences
Tehran, Iran

Hossein Esfandiari
esfandiari@google.com
Google Research
London, United Kingdom

Alireza Rezaeimoghadam
alireza.moghadam@sharif.edu
Sharif University of Technology
Tehran, Iran

Abstract

We provide the first anonymity-preserving algorithm for a centralized decision maker in linear bandit-based multi-user systems. Our algorithm employs successive elimination techniques for linear bandits to build an assignment multi-graph (from users to arms) along with a greedy matching algorithm that efficiently allocates the arms to users. We provide lower and upper bounds for this problem, showing that our algorithm is regret optimal up to a $\sqrt{CK}$ factor.

CCS Concepts

• **Theory of computation** → *Sequential decision making*; **Online learning algorithms**; • **Security and privacy** → Data anonymization and sanitization; **Privacy protections**.

Keywords

Anonymity, Linear Bandits, Multi-User Systems

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1 Introduction

Large-scale platforms such as search, recommendation, and content-delivery systems repeatedly make decisions for many users and adapt the decisions based on the feedback, motivating bandit and RL methods for personalized, structured decision-making, with stochastic linear bandits as a standard model [1, 3, 12]. An important line of research in this domain considers multi-user and collaborative bandits, where information is shared across users to accelerate learning. Examples include distributed and networked bandits [2, 7, 9] and models that exploit latent user structure through clustering or low-dimensional representations [6, 8, 11]. These formulations almost always assume access to each user's individual

reward, and the challenge is how to pool information efficiently across users.

In terms of privacy, several works on *single-user* bandits focus on differential privacy, where individual feedback is observed but suitably perturbed [13, 15]. By contrast, the anonymous bandit framework of Esfandiari et al. [5] enforces privacy for multi-user systems by restricting the observation interface itself: in their anonymous multi-user multi-armed bandit model, the platform never observes user-level rewards, but instead sees only aggregated rewards over groups of size at least C , in the spirit of k -anonymity [14].

In this work, we study this anonymity–utility trade-off in the context of *stochastic linear bandits* for multi-user systems. We consider a centralized learner that repeatedly assigns arms from a common action set $\mathcal{A} \subset \mathbb{R}^d$ to N users over a horizon T . Each user i is characterized by an unknown parameter vector θ_i^* , and the reward obtained when user i plays arm a is a noisy linear function $\langle a, \theta_i^* \rangle$. **Our main contributions are as follows.** We provide the first anonymity preserving algorithm for multi-user stochastic linear bandits and prove a regret bound closely related to those of the work of Esfandiari et al. [5]. To complement our results we build instances that show the regret optimality of our algorithm up to a $\tilde{O}(\sqrt{CK})$ multiplicative factor.

2 Model and Notations

In this section we introduce our problem formulation and notations.

2.1 Stochastic Linear Bandits

We say a random variable X is σ^2 -subgaussian if the mean-normalized variable $Y = X - \mathbb{E}[X]$ satisfies $\mathbb{E}[\exp(sY)] \leq \exp(\sigma^2 s^2/2)$ for all $s \in \mathbb{R}$. We write $[N]$ shortly for the set $\{1, 2, \dots, N\}$.

Next we define stochastic linear bandits. Over rounds $t = 1, \dots, T$, a learner interacts with an environment by selecting an action $x_t \in \mathcal{A} \subset \mathbb{R}^d$ and observing $r_t(x_t) = \langle x_t, \theta^* \rangle + \eta_t$, where $\theta^* \in \mathbb{R}^d$ is unknown and $(\eta_t)_{t=1}^T$ are independent, mean-zero, 1-subgaussian noises. The goal is to minimize the expected cumulative regret

$$\mathcal{R}_T = \mathbb{E}[R_T] = T \sup_{x \in \mathcal{A}} \langle x, \theta^* \rangle - \sum_{t=1}^T \langle x_t, \theta^* \rangle.$$

2.2 Batched Stochastic Bandits

In the *batched* setting, interaction is restricted to B batches. The horizon is partitioned into disjoint batches $b_1, \dots, b_B$ with sizes D_b and $\sum_{b=1}^B D_b = T$. At the start of batch b , the learner selects a policy π_b using data observed up to the end of batch $b-1$ and then



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applies π_b throughout all rounds in b ; updates occur only at batch boundaries, modeling parallel decision making or communication limits.

R_b is the regret incurred in batch b , so the total regret is $R_T = \sum_{b=1}^B R_b$. We provide our arm-elimination algorithm in a batched setting, making it better for systems with limited update availability.

2.3 Anonymous Linear Bandits: Formulation

We consider the *anonymous multi-user bandit* setting introduced by [5]. There are N users and a common action set $\mathcal{A} \subseteq \mathbb{R}^d$ of K arms that spans $\mathbb{R}^d$ and is bounded (i.e. $\forall a \in \mathcal{A} : |a| \leq 1$). Each round $t = 1, 2, \dots, T$, the learner (platform) selects an arm $a_{t,i} \in \mathcal{A}$ for each user $i \in [N]$. If user i plays arm a in round t , they receive a reward $r_{t,i}(a) = \langle a, \theta_i^* \rangle + \eta_{t,i}$ with $|r_{t,i}(a_{t,i})| \leq 1$, where $|\theta_i^*| \leq 1$ is the hidden parameter of user i and $\eta_{t,i}$ is a mean-zero 1-subgaussian noise. We assume the noise RVs are independent across t and i . The goal is to minimize the expected cumulative regret across all users:

$$\begin{aligned} \mathcal{R}_T &= \mathbb{E} \left[\sum_{i=1}^N R_T^{(i)} \right] = \sum_{t=1}^T \left[\sum_{i=1}^N \langle a_i^*, \theta_i^* \rangle - \sum_{i=1}^N \langle a_{t,i}, \theta_i^* \rangle \right] \\ &= \sum_{t=1}^T \sum_{i=1}^N (\langle a_i^*, \theta_i^* \rangle - \langle a_{t,i}, \theta_i^* \rangle) \end{aligned}$$

where $a_i^* = \arg \max_{a \in \mathcal{A}} \langle a, \theta_i^* \rangle$ is the optimal arm for user i and $R_T^{(i)}$ is the regret of user i . The key distinction in this model is that feedback is *anonymous* and *aggregated*. At round t , the algorithm may partition a subset U_t of users into groups $\mathcal{G}_t$ with $|G| \geq C$ for all $G \in \mathcal{G}_t$. All users in a group receive the same arm, and only the group sum is observed (no feedback for users outside U_t)

$$Y_t(G) = \sum_{i \in G} r_{t,i} \quad (G \in \mathcal{G}_t).$$

The grouping pattern $\mathcal{G}_t$ may change each round, but every observed statistic must aggregate at least C users. This C -anonymity constraint makes individual outcomes indistinguishable in the feedback. The following standard assumption is imposed in our work (see [5]):

ASSUMPTION 1. For every arm $a \in \mathcal{A}$, we assume there are at least $U \geq C + 1$ users for whom, a is the optimal arm.

3 Preliminaries for Anonymous Bandit Learning

Our Algorithm has a feedback eliciting component to provide our anonymity guarantees. We will explain this component separately.

3.1 Anonymous decomposition of bipartite graphs

Consider a weighted bipartite graph represented by a non-negative matrix $W = (w_{ij}) \in \mathbb{R}_+^{N \times K}$, where w_{ij} is the demand of user i for arm j . We assume $w_{ij} > 0$ for at least one arm j per user i .

A C -anonymous decomposition of W is a collection of R assignments (matchings) $M_1, \dots, M_R : [N] \rightarrow [K]$ such that:

- (1) Users are only assigned to arms for which they have positive demand, i.e., if $M_r(i) = j$, then $w_{ij} > 0$.

- (2) For each user-arm pair (i, j) , there exist at least w_{ij} assignments M_r where i is matched to j and the number of users assigned to j in M_r is at least $C + 1$. Such assignments are called *informative* for (i, j) .

We focus on *U-batched graphs*, where user i has a demand subset $A_i \subseteq [K]$, with demand $w_{ij} = \frac{D}{|A_i|}$ if $j \in A_i$ and zero otherwise, and each arm j has at least U users interested in it. An *anonymous decomposition algorithm* takes a U-batched graph and outputs a C -anonymous decomposition with at most $\alpha(C, U) \times D + O(NK)$ total assignments, where $D = \sum_{i,j} w_{ij}$ is the total demand. We use the greedy method of [5] that constructs a decomposition with $\alpha(C, U) = K$.

3.2 Feedback eliciting sub-algorithm

We employ the same feedback eliciting in [5]. Let $\pi : [N] \rightarrow [K]$ be a matching from users to arms. We show how to recover (in $2C + 2$ rounds) an unbiased estimate of $\mu_{i,\pi(i)}$ for each i such that

$$|\pi^{-1}(\pi(i))| \geq C + 1$$

Fix i with arm $\pi(i)$. Run $2C+2$ rounds:

- (1) Each round assign every user u to arm $\pi(u)$.
- (2) **Round 0:** For each arm j with $|\pi^{-1}(j)| \geq C+1$, partition $\pi^{-1}(j)$ into groups G_s with $C+1 \leq |G_s| \leq 2C+1$ (fix an order in each G_s). Record group sums $P_{s,0} = \sum_{u \in G_s} r_{t,u}$.
- (3) **Rounds $k = 1:2C+1$:** For each s , let
$$G_{s,k} = \begin{cases} G_s \setminus \{k\text{-th user of } G_s\}, & k \leq |G_s|, \\ G_s, & k > |G_s| \end{cases}$$
and record $P_{s,k} = \sum_{u \in G_{s,k}} r_{t,u}$.
- (4) If i is the k -th user of its group G_s , output $\hat{\mu}_{i,\pi(i)} = P_{s,0} - P_{s,k}$.

In the above procedure, $\mathbb{E}[\hat{\mu}_{i,\pi(i)}] = \mu_{i,\pi(i)}$ and $\hat{\mu}_{i,\pi(i)}$ is the sum of at most $4C + 1$ random variables of the same type.

4 Anonymity in Linear Bandits

In this section, we use a batch arm elimination framework and employ the algorithm presented in [4] with its corresponding optimal number of batches along with the greedy matching procedure of [5].

THEOREM 1. Algorithm 1 has an expected regret of $\tilde{O}(NC\sqrt{TdK})$ for Anonymous Stochastic Linear Bandits.

REMARK 1. R denotes a constant that hides the O notation in lemma 1 and the termination happens where $t > T$.

PROOF. First we start by restating the following lemma. It would mark the role of G-optimal design in our algorithm:

LEMMA 1. [4] Consider any finite action set $A \subset \mathbb{R}^d$ that spans $\mathbb{R}^d$, and let $\delta, \epsilon > 0$ be given. For standard 1-subgaussian noise associated with the feedback of every round, there is a procedure, running in time polynomial in $|\mathcal{A}|$, that produces a multi-set of actions of size $O\left(\frac{d \log(1/\delta)}{\epsilon^2}\right)$ (with possible repetitions) for which the actions themselves span $\mathbb{R}^d$, and when performed in a batched stochastic linear bandit setting, the resulting least-squares estimator $\hat{\theta}$ for θ^* satisfies

$$|\langle \hat{\theta} - \theta^*, a \rangle| \leq \epsilon$$

for every $a \in \mathcal{A}$ with probability at least $1 - \delta$.

Algorithm 1 Anonymous Batch Arm Elimination For Linear Bandits

```

1: Input:  $C, T, K, \mathcal{A}, d$ 
2: Let  $T' = \frac{T}{R(2C+2)K}$ ,  $q = (T')^{\frac{1}{\log T'}}$ ,  $\delta = \frac{1}{KNT^2}$ ,  $b = 1$ .
3:  $\mathcal{A}_{1,1}, \mathcal{A}_{2,1}, \dots, \mathcal{A}_{N,1} = \mathcal{A}$ 
4: while  $b \leq \lceil \log(T') \rceil$  do
5:    $\epsilon_b = \sqrt{\frac{(4C+1)d \log(KNT^2)}{q^b}}$ ,  $D_b = (2C+2)KR \times q^b$ 
6:   for  $i = 1, 2, \dots, N$  do
7:      $S_i \leftarrow$  multi-set given by approximate G-optimal Design
       with input  $\epsilon_b, \delta, \mathcal{A}_{i,b}$  provided in lemma 1
8:      $S_i(j)$  denotes the repeat times of arm  $j$  in the multi-set  $S_i$ 
9:   end for
10:   $\forall j \in [K]$  denote  $P_j : [N] \rightarrow [K]$  as the following matching:
11:     $P_j(i) \leftarrow \begin{cases} j & j \in \mathcal{A}_{i,b} \\ \text{UNIFORM}(\mathcal{A}_{i,b}) & \text{otherwise} \end{cases}$ 
12:  Play every matching  $P_j$  a total of  $(2C+2) \times \max_i S_i(j) \leq$ 
    $(2C+2)Rq^b$  times, doing the feedback eliciting subroutine
    $\max_i S_i(j)$  times in total for each  $P_j$ 
13:  For the remaining rounds of batch, play an arbitrary arm in
   the active set of arms of each user
14:  Update the parameter's estimation for every user using the
   least squares estimator to obtain  $\hat{\theta}_{i,b}$ 
15:   $\forall i \in [N] : \mathcal{A}_{i,b+1} \leftarrow \{a \in \mathcal{A}_{i,b} \mid \langle a, \hat{\theta}_{i,b} \rangle \geq$ 
    $\max_{a' \in \mathcal{A}_{i,b}} \langle a', \hat{\theta}_{i,b} \rangle - 2\epsilon_b\}$ ,  $b \leftarrow b+1$ 
16: end while

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If the noises are σ^2 -subgaussian random variable with $\sigma^2 \geq 1$, the same claim holds but with a sample complexity of $O(\frac{\sigma^2 d \log(1/\delta)}{\epsilon^2})$.

REMARK 2. Here, because of the combinatorial complexity of the feedback eliciting subroutine, the noises are not the standard 1-subgaussian and they are $4C+1$ -subgaussian.

PROOF. We prove this lemma for the more general case of having σ^2 -subgaussian noise. The argument is similar to the argument presented in [10]. We first provide and prove the key lemma:

LEMMA 2. Let $\eta_1, \dots, \eta_n$ be a sequence of independent C -subgaussian random variables and $a_1, \dots, a_n \in \mathbb{R}^d$ be a fixed sequence with $\text{span}(a_1, \dots, a_n) = \mathbb{R}^d$, and let $X_1, \dots, X_n$ be given by

$$X_t = \langle \theta_*, a_t \rangle + \eta_t$$

for some $\theta_* \in \mathbb{R}^d$. The least squares estimator of θ_* is

$$\hat{\theta} = V^{-1} \sum_{t=1}^n a_t X_t, \quad V = \sum_{t=1}^n a_t a_t^\top.$$

and we can obtain the following concentration inequality:

$$\mathbb{P}\left(\langle \hat{\theta} - \theta_*, a \rangle \geq \sqrt{2 \sigma^2 \|a\|_{V^{-1}}^2 \log\left(\frac{1}{\delta}\right)}\right) \leq \delta.$$

Now we could use this lemma along with Approximate G-optimal Design to obtain an algorithm that has the desired sample complexity. Here we repeat the argument presented in **Section 21.1** [10].

Let $\pi : \mathcal{A} \rightarrow [0, 1]$ be a distribution on $\mathcal{A}$ ($\sum_{a \in \mathcal{A}} \pi(a) = 1$) and

$$V(\pi) = \sum_{a \in \mathcal{A}} \pi(a) a a^\top, \quad g(\pi) = \max_{a \in \mathcal{A}} \|a\|_{V(\pi)^{-1}}^2.$$

In optimal experimental design, the distribution π is called a *design*, and minimizing g is the *G-optimal design problem*. Suppose a design π is given. For any $a \in \mathcal{A}$ set

$$n_a = \begin{cases} \left\lceil \frac{2\sigma^2 \pi(a) g(\pi)}{\epsilon^2} \log\left(\frac{1}{\delta}\right) \right\rceil & a \in \text{Supp}(\pi) \\ 0 & \text{otherwise} \end{cases}$$

Here, we choose action $a \in \mathcal{A}$ exactly n_a times ensuring

$$V = \sum_{a \in \text{Supp}(\pi)} n_a a a^\top \succeq \frac{2\sigma^2 g(\pi)}{\epsilon^2} \log\left(\frac{1}{\delta}\right) V(\pi),$$

Inverting the matrix inequality above we can write for any $a \in \mathcal{A}$

$$\begin{aligned} \|a\|_{V^{-1}}^2 &= a^\top V^{-1} a \leq \frac{\epsilon^2}{2\sigma^2 g(\pi) \log\left(\frac{1}{\delta}\right)} a^\top V(\pi)^{-1} a \\ &\leq \frac{\epsilon^2}{2\sigma^2 \log\left(\frac{1}{\delta}\right)}, \end{aligned}$$

Meaning that by lemma 2, for any $a \in \mathcal{A}$, with probability $1 - \delta$,

$$\langle \hat{\theta} - \theta_*, a \rangle \leq \sqrt{2\sigma^2 \|a\|_{V^{-1}}^2 \log\left(\frac{1}{\delta}\right)} \leq \epsilon.$$

The number of times we pulled an action to ensure an accuracy width of no more than ϵ is bounded by

$$\begin{aligned} n &= \sum_{a \in \text{Supp}(\pi)} n_a = \sum_{a \in \text{Supp}(\pi)} \left\lceil \frac{2\sigma^2 \pi(a) g(\pi)}{\epsilon^2} \log\left(\frac{1}{\delta}\right) \right\rceil \\ &\leq |\text{Supp}(\pi)| + \frac{2\sigma^2 g(\pi)}{\epsilon^2} \log\left(\frac{1}{\delta}\right). \end{aligned}$$

Note that with G-optimal design we can identify a policy π^* with $g(\pi^*) = d$ with a cardinality of at most $O(d^2)$ as stated in **Theorem 21.1** [10]. Also note that an exact design is not needed to obtain a good regret in terms of order and approximate G-optimal design could be employed to identify a policy that has $g(\pi) = O(d)$. $\square$

We denote the following event for each batch and note that it occurs with a desired high probability because of our design:

For every user $i \in [N]$, batch b , and arm in the active set of arms of the user i in the beginning of the batch b ($a \in \mathcal{A}_{i,b}$), $|\langle a, \hat{\theta}_{i,b} - \theta_i \rangle| \leq \epsilon_b$ holds at the end of batch b . Note that there are N users, $\leq T$ batches, K arms so by union bound and the proposed G-optimal Design in lemma 1, the event of this inequality being held for every arm, batch and user holds with probability at least $1 - \delta \times KNT = 1 - \frac{\delta}{T}$. To bound the expected regret, we condition on whether this event holds, a linear regret of NT suffices if it does not since the associated probability is negligible. If this event holds:

First note that under this event, the best arm will not be eliminated during the procedure because of the design in the elimination rule. Suppose that we are in batch b and the best arm is not eliminated up to now. Denote the best arm for user i as a_i^* and the best estimated arm for user i up to batch b as $\hat{a}_{i,b}^*$. Because of

$$\langle a_i^*, \hat{\theta}_{i,b} \rangle \geq \langle a_i^*, \theta_i \rangle - \epsilon_b \geq \langle \hat{a}_{i,b}^*, \theta_i \rangle - \epsilon_b \geq \langle \hat{a}_{i,b}^*, \hat{\theta}_{i,b} \rangle - 2\epsilon_b,$$

the best arm will not be eliminated in the elimination phase of all batches so $\forall a \in \mathcal{A}_{i,b+1}$ we have that a regret of at most $4\epsilon_b$:

$$\begin{aligned} \max_{a \in \mathcal{A}} \langle a, \theta_i \rangle &= \max_{a \in \mathcal{A}_{i,b+1}} \langle a, \theta_i \rangle \leq \langle a, \hat{\theta}_{i,b} \rangle + \epsilon_b \\ &\leq \max_{a \in \mathcal{A}_{i,b+1}} \langle a, \hat{\theta}_{i,b} \rangle + \epsilon_b \leq \langle a, \hat{\theta}_{i,b} \rangle + 3\epsilon_b \leq \langle a, \theta_i \rangle + 4\epsilon_b, \end{aligned}$$

where the equality follows because the best arm will not be eliminated. The first and fourth inequality follow since the event holds, $|\langle a, \hat{\theta}_{i,b} - \theta_i \rangle| \leq \epsilon_b$. The third inequality follows because for every active arm $a \in \mathcal{A}_{i,b+1}$ we have $\langle a, \hat{\theta}_{i,b} \rangle \geq \max_{a' \in \mathcal{A}_{i,b}} \langle a', \hat{\theta}_{i,b} \rangle - 2\epsilon_b$.

Note that the term $4C+1$ in ϵ_b is because of the $4C+1$ -subgaussian noise of the feedback eliciting subroutine as stated in [5]. In particular, during every batch, we use the implications of lemma 1 with the correct noise which is $\sigma^2 = 4C + 1$. Now the regret for each user i , by using a worst-case upper bound for the first batch and using the bound obtained above for all the other batches, could be written as:

$$\begin{aligned} R_T^{(i)} &\leq D_1 + \sum_{b=2}^{\log T'} 4D_b \epsilon_{b-1} \leq q(2C+2)KR + \sum_b 4\epsilon_b q^{b+1} (2C+2)KR \\ &\leq (2C+2)KR(q + 4 \sum_b \frac{\sqrt{(4C+1)d \log(KNT^2)}}{q^{b/2}} q^{b+1}) \\ &\leq O(RC^{3/2} K \sqrt{d \log(KNT^2)} q \sum_b q^{b/2}) \\ &= O(C^{3/2} K \sqrt{d \log(KNT^2)} q^{\log T'/2}) \\ &= O(C^{3/2} K \sqrt{dT' \log(KNT^2)}) = O(C \sqrt{dKT \log(KNT)}) \end{aligned}$$

and by summing the regret of each user, we are done. $\square$

Finally we present a lower bound which matches our upper bound up to $\sqrt{CK}$ factor. Similar to [5] the idea is to investigate this problem when the learning procedure for each user is independent of the others and there is no centralized learner.

THEOREM 2. *There is a finite-armed environment with $|\mathcal{A}| = 2^d$ in which for every ordered-set of policies $\Pi = (\pi_1, \pi_2, \dots, \pi_N)$ for users there exist hidden vectors $\Theta = (\theta_1, \theta_2, \dots, \theta_N)$ such that:*

$$\mathcal{R}(\Pi, \Theta) \geq \frac{\exp(-2)}{8} Nd\sqrt{CT}.$$

PROOF. The proof idea is similar to that of [5]. Building the same environment, we assume that the number of users, the anonymity parameter, the time horizon and the dimension of action set are denoted by N, C, T, d respectively. The provided environment has gaussian standard noise. We will tell user i , the hidden parameters of all the other users. This means that the mean reward of all arms of the other users is known to him. He will also realize the arms played by all users after the round. He can use this information to build an unbiased estimator for himself at each timestep. In particular, if user i selects arm $\pi_i(t)$ at time t and $G_t(i)$ denotes the set of all the other users in his group, he will realize $r_{t,i}(\pi_i(t)) + \sum_{u \in G_t(i)} r_{t,u}(\pi_u(t))$ but since (after the round) he knows the arms played by users in $G_t(i)$, i.e. $\pi_u(t) \forall u \in G_t(i)$, along with the hidden parameter of other users, he can calculate and subtract the corresponding means of other users actions to obtain an unbiased feedback for his action. Note that feedback is only available to groups with cardinality greater than or equal to C , meaning that when a feedback is available to the user, its noise would be the sum of at least C

standard normal RVs which would have a variance of C . Moreover, in this instance of the game, we view users as independent learners with additional information rather than having a centralized learner for all users. Now we just require a lower bound for single-user finite-armed stochastic linear bandits:

LEMMA 3. [10] Theorem 24.1. *Let $\mathcal{A} = \{-1, 1\}^d$ and $\Theta = \{-T^{-1/2}, T^{-1/2}\}^d$. Then for any policy π , there exists a vector $\theta \in \Theta$ such that:*

$$\mathcal{R}(\pi, \theta) \geq \frac{\exp(-2)}{8} d\sqrt{T}$$

REMARK 3. *In the more general case of variance C (instead of 1), the bound will change to $\frac{\exp(-2)}{8} d\sqrt{CT}$.*

The action set of all users is denoted by $\mathcal{A} = \{-1, 1\}^d$. By lemma 3, For every user, we can say that there is a single parameter θ_i^* for which the lower bound holds. So there exists a sequence of parameters for which the corresponding user will suffer the regret lower bound and since the Cumulative regret is the sum of regrets of each user, we have proved a worst-case lower bound of $\frac{\exp(-2)}{8} Nd\sqrt{CT}$. $\square$

References

- [1] Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. 2011. Improved algorithms for linear stochastic bandits. *Advances in Neural Information Processing Systems* 24 (2011).
- [2] Swapna Bucapatnam, Jian Tan, and Li Zhang. 2015. Information sharing in distributed stochastic bandits. In *2015 IEEE Conference on Computer Communications (INFOCOM)*. IEEE, 2605–2613.
- [3] Varsha Dani, Thomas P Hayes, and Sham M Kakade. 2008. Stochastic linear optimization under bandit feedback.
- [4] Hossein Esfandiari, Amin Karbasi, Abbas Mehrabian, and Vahab Mirrokni. 2021. Regret Bounds for Batched Bandits. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. AAAI Press, 7340–7348. doi:10.1609/aaai.v35i8.16872
- [5] Hossein Esfandiari, Vahab Mirrokni, and Jon Schneider. 2022. Anonymous bandits for multi-user systems. *Advances in Neural Information Processing Systems* 35 (2022), 12422–12434.
- [6] Claudio Gentile, Shuai Li, Purushottam Kar, Alexandros Karatzoglou, Giovanni Zappella, and Evans Etrue. 2017. On context-dependent clustering of bandits. In *International Conference on machine learning*. PMLR, 1253–1262.
- [7] Ravi Kumar Kolla, Krishna Jagannathan, and Aditya Gopalan. 2018. Collaborative learning of stochastic bandits over a social network. *IEEE/ACM Transactions on Networking* 26, 4 (2018), 1782–1795.
- [8] Nathan Korda, Balazs Szorenyi, and Shuai Li. 2016. Distributed clustering of linear bandits in peer to peer networks. In *International conference on machine learning*. PMLR, 1301–1309.
- [9] Peter Landgren, Vaibhav Srivastava, and Naomi Ehrlich Leonard. 2016. Distributed cooperative decision-making in multiarmed bandits: Frequentist and bayesian algorithms. In *2016 IEEE 55th Conference on Decision and Control (CDC)*. IEEE, 167–172.
- [10] Tor Lattimore and Csaba Szepesvári. 2020. *Bandit algorithms*. Cambridge University Press.
- [11] Odalric-Ambrym Maillard and Shie Mannor. 2014. Latent bandits. In *International Conference on Machine Learning*. PMLR, 136–144.
- [12] Paati Rusmevichientong and John N Tsitsiklis. 2010. Linearly parameterized bandits. *Mathematics of Operations Research* 35, 2 (2010), 395–411.
- [13] Roshan Shariff and Or Sheffet. 2018. Differentially private contextual linear bandits. *Advances in Neural Information Processing Systems* 31 (2018).
- [14] Latanya Sweeney. 2002. k-anonymity: A model for protecting privacy. *International journal of uncertainty, fuzziness and knowledge-based systems* 10, 05 (2002), 557–570.
- [15] Aristide C Tossou and Christos Dimitrakakis. 2016. Algorithms for differentially private multi-armed bandits. In *Thirtieth AAAI Conference on Artificial Intelligence*.